Exercise 5: Task Management System

CODE:

```
class Task {
    int taskId;
    String taskName;
    String status;
    Task next;
    Task(int taskId, String taskName, String status) {
        this.taskId = taskId;
        this.taskName = taskName;
        this.status = status;
        this.next = null;
    }
}
public class Main {
    static Task head = null;
    // Add Task
    static void addTask(int id, String name, String status) {
        Task newTask = new Task(id, name, status);
        if (head == null) {
            head = newTask;
        } else {
            Task temp = head;
            while (temp.next != null) {
                temp = temp.next;
            }
            temp.next = newTask;
        }
    }
```

```
// Search Task
static void searchTask(int id) {
    Task temp = head;
    while (temp != null) {
        if (temp.taskId == id) {
            System.out.println("\nTask Found");
            System.out.println("ID\tTask Name\tStatus");
            System.out.println(temp.taskId + "\t" + temp.taskName + "\t\t" + temp.status);
            return;
        }
        temp = temp.next;
    }
    System.out.println("\nTask not found.");
}
// Traverse Tasks
static void traverseTasks() {
    System.out.println("ID\tTask Name\tStatus");
    Task temp = head;
    while (temp != null) {
        System.out.println(temp.taskId + "\t" + temp.taskName + "\t\t" + temp.status);
        temp = temp.next;
    }
}
// Delete Task
static void deleteTask(int id) {
    if (head == null)
        return;
    if (head.taskId == id) {
        head = head.next;
```

```
            return;
        }
        Task current = head;
        while (current.next != null && current.next.taskId != id) {
            current = current.next;
        }
        if (current.next != null) {
            current.next = current.next.next;
        }
    }
    public static void main(String[] args) {
        addTask(101, "Design", "Pending");
        addTask(102, "Coding", "In Progress");
        addTask(103, "Testing", "Pending");
        System.out.println("Task List");
        traverseTasks();
        searchTask(102);
        deleteTask(103);
        System.out.println("\nTask List After Deletion");
        traverseTasks();
    }
}
```

OUTPUT:

```
Task List
ID  Task Name   Status
101 Design      Pending
102 Coding      In Progress
103 Testing     Pending

Task Found
ID  Task Name   Status
102 Coding      In Progress

Task List After Deletion
ID  Task Name   Status
101 Design      Pending
102 Coding      In Progress
```